



CANDIDATE
NAME

ANSWER KEY

CIVICS
GROUP

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REGISTRATION
NUMBER

H2 Biology

9744/03

Paper 3 Long Structured and Free-response Questions

20 September 2021

2 hours

Candidates answer **Section A** on the Question Paper and **Section B** on the separate Answer Booklet.

Additional Materials: 12-page Answer Booklet.

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Section A

Answer **all** questions.

Section B

Answer **one** question.

Write your answer to each part of the question on a fresh sheet of paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
Section B	
4 OR 5	
Total	75

This document consists of **25** printed pages.

Section A

Answer **all** the questions in this section.

- 1** Fig. 1.1 shows the location of two genes, *HFE* and *EPM2A*, on human chromosome 6.

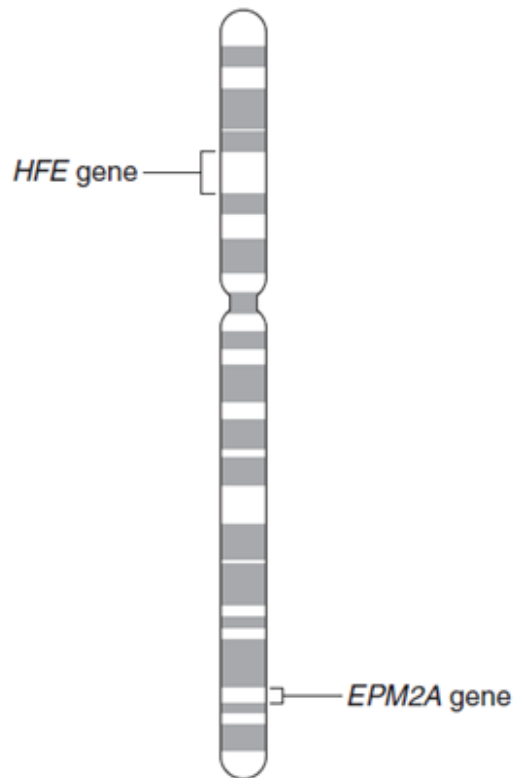


Fig. 1.1

The protein product of the *HFE* gene is involved in the absorption of iron. A mutation of the *HFE* gene, where a cysteine is replaced by tyrosine at position 282, can lead to the disorder hereditary haemochromatosis.

Laforin, the protein product of the *EPM2A* gene, has a number of roles in the body. A deletion mutation of the *EPM2A* gene has been linked to Lafora disease.

Fig. 1.2 shows the genotype of a healthy human female, Mary.

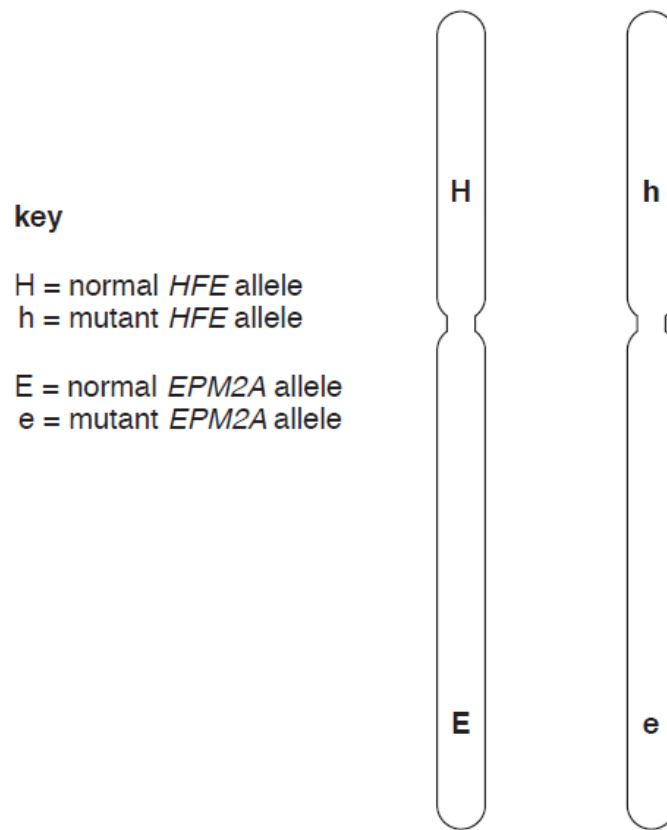
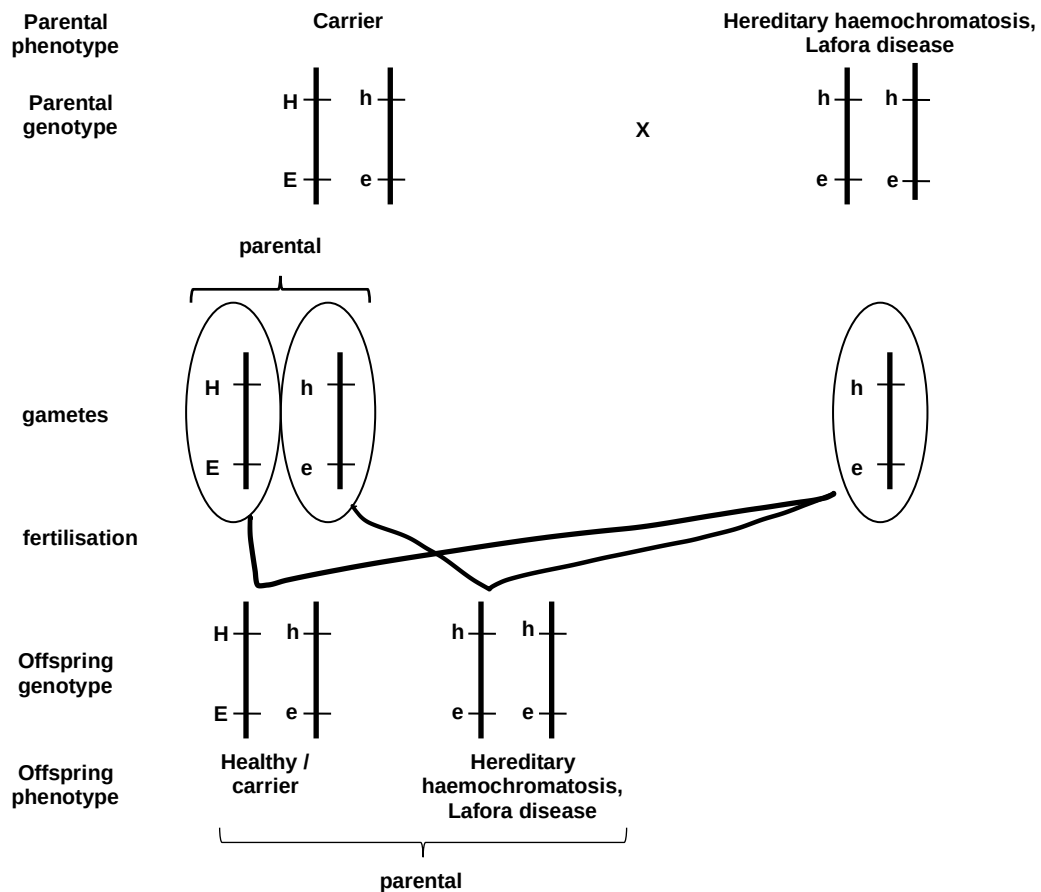


Fig. 1.2

Mary's husband has both hereditary haemochromatosis and Lafora disease. They have two children:

- a daughter, who is healthy
- a son, who has both hereditary haemochromatosis and Lafora disease.

(a) Draw a genetic diagram to explain the cross between Mary and her husband described above.



[1] parental phenotype + genotype

[1] gametes (circled)

[1] fertilisation

[1] "parental" types indicated

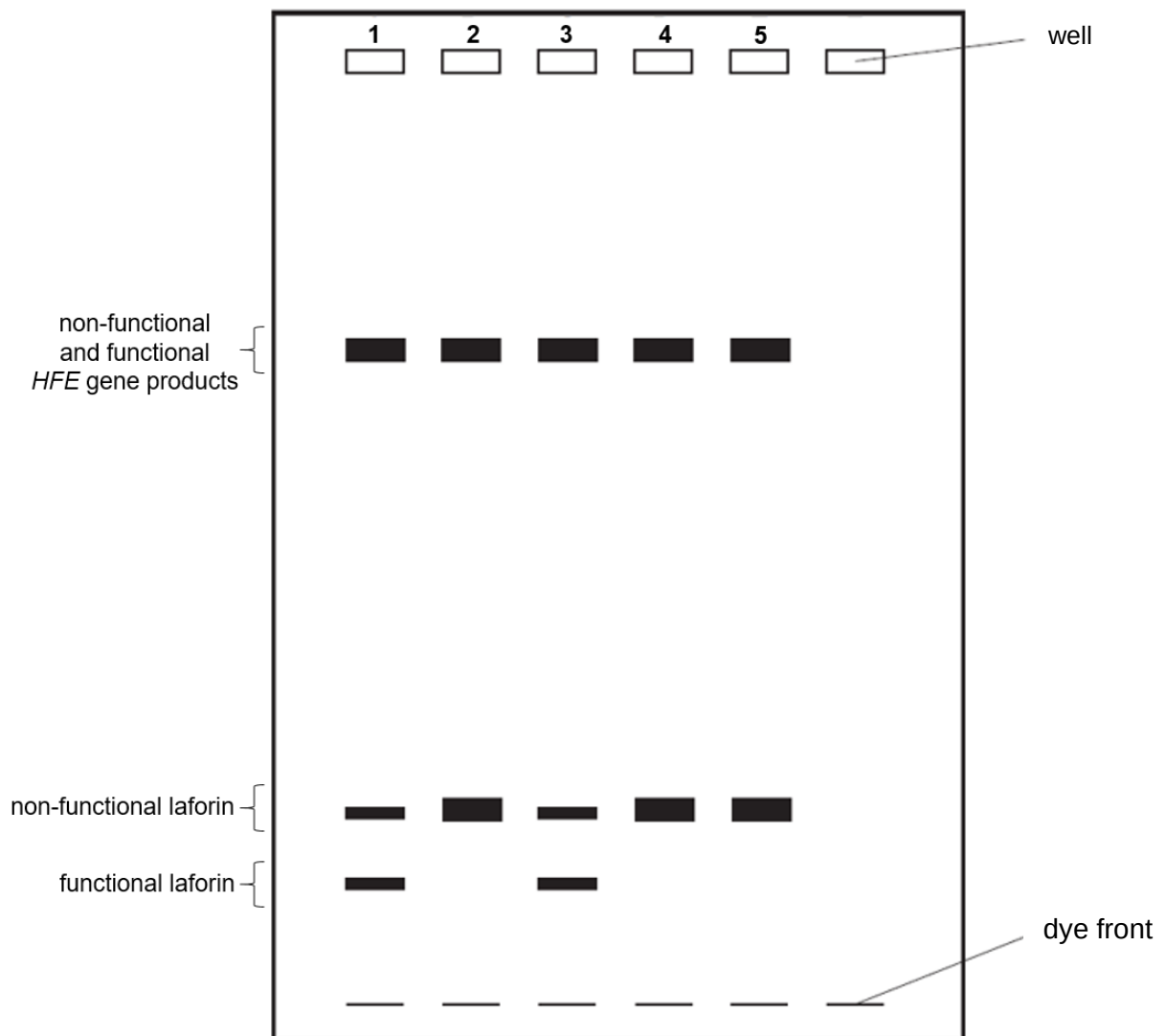
[1] genotype correctly linked to phenotype for offspring

The technique of polyacrylamide gel electrophoresis (PAGE) is used to separate and identify proteins. One method of PAGE involves treating proteins with a detergent, such as sodium dodecyl sulfate (SDS), to dissociate proteins into their constituent polypeptide subunits. Proteins treated with SDS have a uniform net charge on each polypeptide so that during electrophoresis they are separated only on the basis of their relative molecular mass.

After treatment with SDS, proteins are placed in wells cut into a polyacrylamide gel. A dye is added to each sample to show the progress of the samples across the gel. A current is applied to the gel and when the dye reaches a point towards the end of the gel, the current is switched off. The relative mobility of each polypeptide is calculated as follows:

$$\frac{\text{distance travelled by polypeptide band}}{\text{distance travelled by dye front}}$$

Gene products of the *HFE* gene and *EPM2A* gene from Mary and her family were analysed with SDS-PAGE and the results are shown in Fig. 1.3.



Legend:

- Lane 1 – Mary
- Lane 2 – Mary's husband
- Lane 3 – Mary's daughter
- Lane 4 – Mary's son
- Lane 5 – Individual X

Fig. 1.3

- (b) (i) The distance travelled by the polypeptide band of both the non-functional and functional *HFE* gene products is 38 mm, while that travelled by the dye front is 128 mm. The relative mobility of non-functional and functional *HFE* gene products is 0.297.

Calculate the relative mobility of non-functional and functional laforin, and fill in Table 1.1 with your calculated values.

Table 1.1

protein	relative molecular mass	relative mobility
non-functional <i>HFE</i> gene product	68000	0.297
functional <i>HFE</i> gene product	68000	0.297
non-functional laforin	34000	0.797
functional laforin	29000	0.875

[1]

- (ii) Using the information provided, explain the relative mobility of the non-functional and functional *HFE* gene products.

.....

 [2]

1. Non-functional and functional *HFE* gene products have the same relative molecular mass of 68000 and will have the same relative mobility of 0.304;
2. Because only one amino acid was changed (Cys → Tyr), and hence, they have the same number of amino acids;

- (iii) Individual X is the third child of Mary and her husband, and suffers from Lafora disease but not hereditary haemochromatosis. Fig. 1.4 shows a pair of chromosome 6 in individual X.

On Fig. 1.4, indicate the genotype of individual X in the boxes.

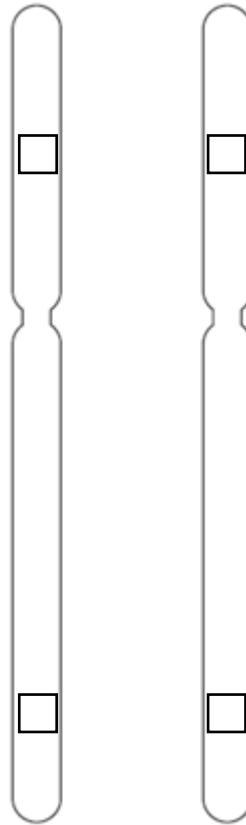


Fig. 1.4

- allele H and allele e on one chromosome
- allele h and allele e on one chromosome

[1]

- (iv) On Fig. 1.4, circle the chromosome that individual X inherited from Mary.

[1]

Chromosome with allele H and allele e.

- (v) Using your knowledge on gamete formation, describe how individual X came to inherit the combination of alleles indicated in Fig. 1.4 from Mary.

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- [2]
1. During prophase I of meiosis, chiasma formed at the region between HFE gene locus and EPM2A gene locus;
 2. Crossing over then took place between non-sister chromatids of Mary's pair of chromosome 6;

This resulted in alleles H and e being on the same chromosome 6 and passed on to individual X from Mary;

- (c) Suggest if genetic diseases such as Lafora disease is an example of continuous or discontinuous variation.

Provide **two** reasons to support your answer.

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.....

.....[3]

1. Discontinuous variation;

(any 2 below)

2. Distinct and discrete phenotypes, either have Lafora disease or healthy;
3. Only one gene, the EPM2A gene determines the trait;
4. The trait is solely genetically determined, and not affected by environmental factors;

- (d) Mary and her family reside in Australia where the giant moray eel can be found. The sizes of eel populations tend to remain relatively stable despite eels producing large numbers of offspring.

Suggest **two** reasons why the population sizes of eels tend to remain relatively stable.

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..... [2]

1. Predation / fishing;
2. Limited food resources / competition for food;
3. Disease;

The generation time of a species is the average number of years between the birth of one generation (parents) and the birth of the next generation (offspring). Fig. 1.5 shows the generation time of the American cockroach and the African elephant.

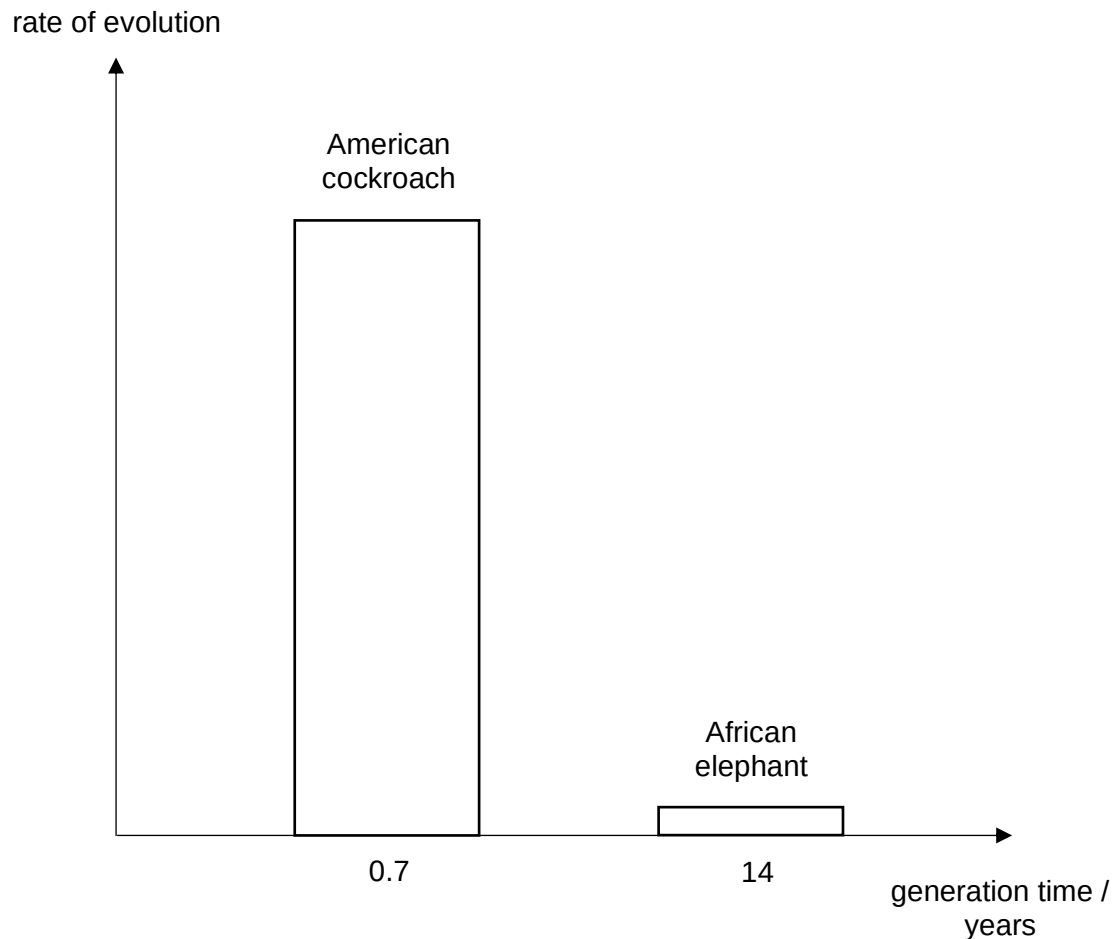


Fig. 1.5

- (e) One of the most important factors in determining the rate of evolution is the introduction of new alleles within a species.

Describe and explain the relationship shown in Fig. 1.5.

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-[3]
1. As the generation time increases, the rate of evolution decreases;
 2. Mutation results in the formation of new alleles;
 3. With a shorter generation time, the new alleles are passed on to the next generation at a quicker rate, and the rate of change in allele frequency increases;

The habitat of the giant moray eel includes coral reefs as shown in Fig. 1.6.



Fig. 1.6

Corals grow in shallow sea water. Corals consist of colonies of small animals called polyps. These polyps have photosynthetic algae living within their cells, and this relationship is advantageous to both the coral polyps and to the algae.

The algae that live within the cells of coral polyps can also live independently as free-living algae in the sea water.

- (f) Suggest why the rate of photosynthesis of algae that live within the cells of coral polyps is higher than that of free-living algae.

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.....[3]

1. Concentration of CO₂ is higher within the cells of the polyps than sea water
OR
Concentration of CO₂ is a limiting factor in sea water but not within the cells of the polyps;
2. This is because CO₂ is produced during the respiration in the cells of the polyps;
3. In the presence of higher CO₂ concentration, rate of Calvin cycle / light-independent stage is higher in the algae that live within the cells of coral polyps, leading to a higher rate of photosynthesis;

The algae that live within the cells of coral polyps have five different chloroplast pigments.

Table 1.2 shows the wavelengths of light with the largest absorption for each algae chloroplast pigment.

Table 1.2

chloroplast pigment	wavelength / nm
chlorophyll a	430
peridinin	456
chlorophyll c ₂	450
dinoxanthin	442
β-carotene	454

Corals can be kept in glass tanks that are usually lit by lamps radiating mainly violet and blue light, with wavelengths in the range of 400 nm to 490 nm.

- (g) (i) With reference to Table 1.2, explain why lamps radiating mainly violet and blue light are expected to increase the rate of photosynthesis in algae more than lamps radiating light of all wavelengths.

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[2]

1. All five pigments show largest peaks of light absorption for wavelengths in the range of 430 nm to 456 nm;
2. When the coral polyps are exposed to lamps radiating violet and blue light, there would be higher rate of photo-activation of P700/P680 in photosystems, and therefore, higher rate of photosynthesis in algae;

- (ii) A high rate of photosynthesis in the algal cells will result in high rate of coral growth.

Explain why.

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[1]

Glucose / oxygen is produced at a higher rate by algae (leading to increase in cellular respiration which produces more ATP), which can then be used by coral polyps, leading to more growth of the coral polyps;

(R: glucose respired by coral polyps to result in more growth without mention of ATP)

Short term exposure to increased temperature can trigger a change in gene expression in corals.

For corals exposed to higher temperature, the expression of some proteins was compared against the control. Corals in the control group were exposed to normal temperature. The difference in gene expression between both groups of corals is shown in Table 1.3.

Table 1.3

protein	average concentration of protein / μM	
	control group	corals exposed to higher temperature
BCL2	16	232
cathepsin	184	10
HSP90	20	149

- (h) (i) Calculate the percentage increase in expression of BCL2 for corals exposed to higher temperature.

[1]

$$\frac{216}{16} * 100\% = 1350\%$$

- (ii) Using Table 1.3, describe the effect of higher temperature on the expression of HSP90.

.....

.....[1]

Higher temperature increases the expression of HSP90, as shown by average concentration of HSP90 increasing from 20 μM to 149 μM for corals exposed to higher temperature;

- (iii) Based on your knowledge of gene expression, suggest how the expression of HSP90 in corals exposed to higher temperature can be regulated at the transcriptional level.

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..... [2]

1. Activator bind to enhancer, which then stabilise the assembly of RNA polymerase and general transcription factors at the promoter to form the transcription initiation complex;
2. Rate of / more HSP90 mRNA are formed, leading to an increased production of HSP90 from translation;

[Total: 30]

- 2 Fig. 2.1 shows some organelles in a mammalian liver cell.

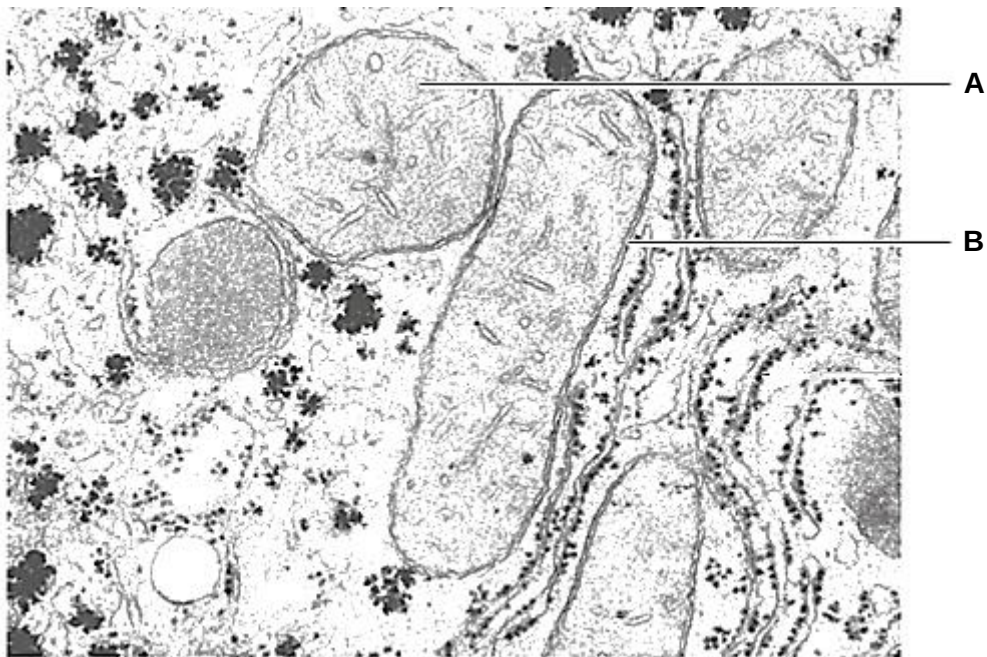


Fig. 2.1

- (a) With reference to Fig. 2.1, explain if **A** and **B** are the same organelle.

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..... [2]

1. Yes, A and B are the same organelle, mitochondria;
2. Both are double membrane bound organelles, with the inner membrane thrown into folds / cristae;

- (b) During maturation, a red blood cell loses its nucleus and most other organelles, such as the ribosomes, rough and smooth endoplasmic reticulum, lysosomes and mitochondria. This gives more space for haemoglobin molecules.

Although there are no mitochondria, a mature red blood cell is still able to synthesise ATP.

Explain how a mature red blood cell can continue to synthesise ATP.

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..... [3]

1. Glycolysis and lactate fermentation (Note to marker → A: anaerobic respiration) occur in the cytosol of the red blood cell;
2. Glycolysis produces 2 net ATP will be produced per glucose molecule;
3. Lactate fermentation regenerates NAD for glycolysis to continue;

(c) A number of chemicals can disrupt respiration. Some of these are listed in Table 2.1.

Table 2.1

chemical	effect on mitochondrion
oligomycin	inhibits ATP synthase, thereby leading to steep proton gradient across the cristae and inhibition of active transport of protons across cristae
FCCP	causes protons to leak out of the intermembrane space back into the mitochondrial matrix without going through ATP synthase
antimycin-A + rotenone	inhibits the electron transport chain

Scientists measured the oxygen consumption rate (OCR) of some cells. They then investigated how the OCR changed when oligomycin was added at 23 minutes, followed by FCCP at 56 minutes, followed by antimycin-A + rotenone at 93 minutes.

Fig. 2.2 shows the results obtained.

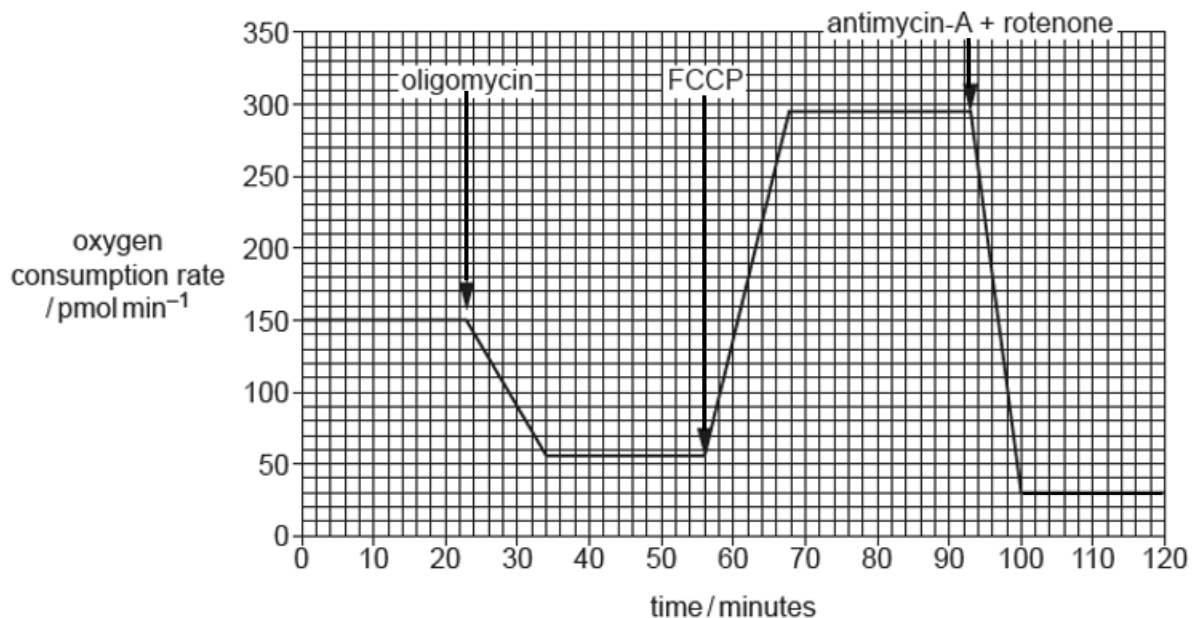


Fig. 2.2

- (i) Explain each of the following two statements with reference to events at the inner mitochondrial membrane.

Oligomycin causes a drop in OCR

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 [2]

1. With active transport of protons inhibited / ETC inhibited, electrons cannot travel down / be passed down the electron transport chain (ETC);
2. Therefore, no electrons are passed to the final electron acceptor, oxygen, and thus OCR drops from 150 pmol min⁻¹ to 57 (+/- 1) pmol min⁻¹;

FCCP causes a rise in OCR

.....

 [2]

1. In the presence of FCCP, proton gradient dissipates / becomes less steep;
2. Electrons will be able to move down the ETC and be passed on to oxygen eventually, thereby increasing OCR from 57 (+/- 1) pmol min⁻¹ to 295 pmol min⁻¹;

- (ii) During respiration, protons 'leak' through the cristae into the mitochondrial matrix. The OCR as a result of this leakage can be calculated by determining the difference between the OCR 20 minutes after the addition of oligomycin and the OCR 20 minutes after the addition of antimycin-A + rotenone.

Calculate the OCR caused by proton leakage. Show your working clearly.

$$57 (+/- 1) - 30 = 26 \text{ pmol min}^{-1}$$

.....pmol min⁻¹

[1]

[Total: 10]

- 3 Down's syndrome may be caused by a trisomy of chromosome 21, but may also result from the fusion of a large portion of chromosome 21 with chromosome 14, forming a single chromosome 14-21. The remaining portion of chromosome 21 is lost, but has no effect on the individual.

Fig. 3.1 shows the karyotype of a diploid ($2n$) testis cell from a phenotypically normal individual carrying a 14-21 fusion chromosome, indicated with an arrow. The centromere of each chromosome is positioned on the dotted line. This individual has a higher chance of having a child with Down's syndrome.

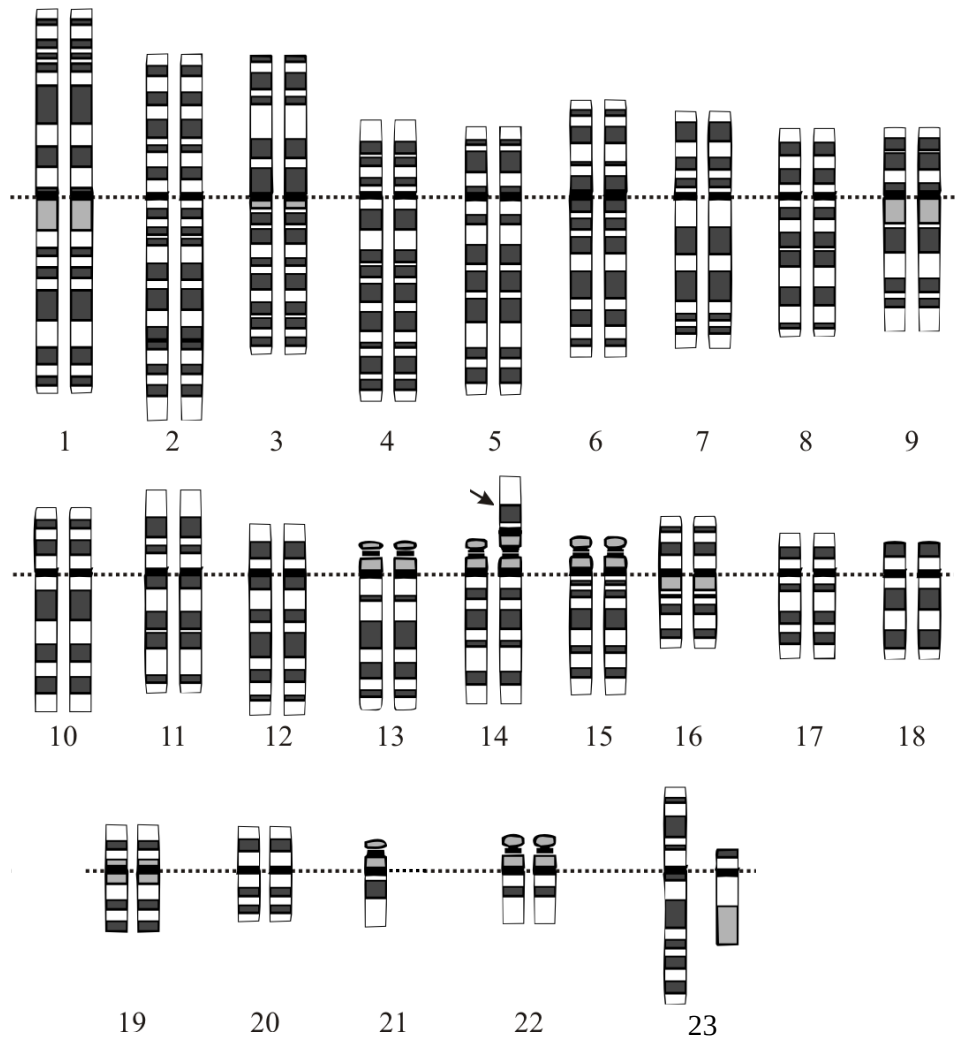


Fig. 3.1

- (a) (i) Suggest the type of chromosomal mutation that has occurred in this individual.

..... [1]
Translocation;

- (ii) State and explain **one other** phenotypic feature of this individual.

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 [1]
 This individual is **male**, as the sex chromosomes are **XY** / sex chromosomes are **not homologous**;

- (b) Fig. 3.2 shows all possible combinations of chromosomes 14, 21, and 14-21 in the gametes produced by the same individual whose karyotype is shown in Fig. 3.1.

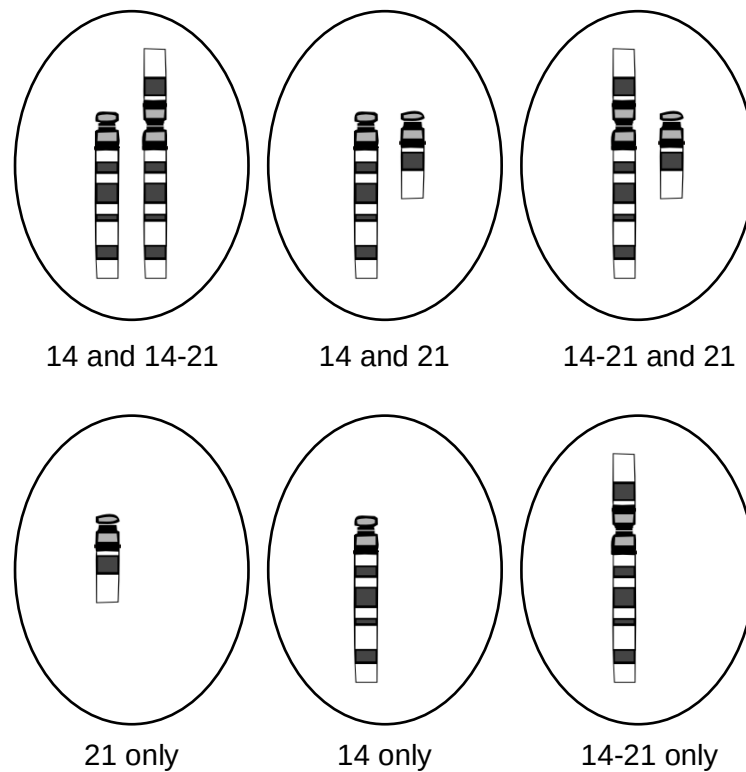


Fig. 3.2

It is known that monosomy-14, monosomy-21, and trisomy-14 are all lethal conditions, and the embryo will not develop beyond 20 weeks.

- (i) Assuming that a new-born child inherits a normal gamete from the other parent, state the probability that the child is phenotypically normal.

probability = **2/3**..... [1]

- (ii) Explain your answer in (b)(i).

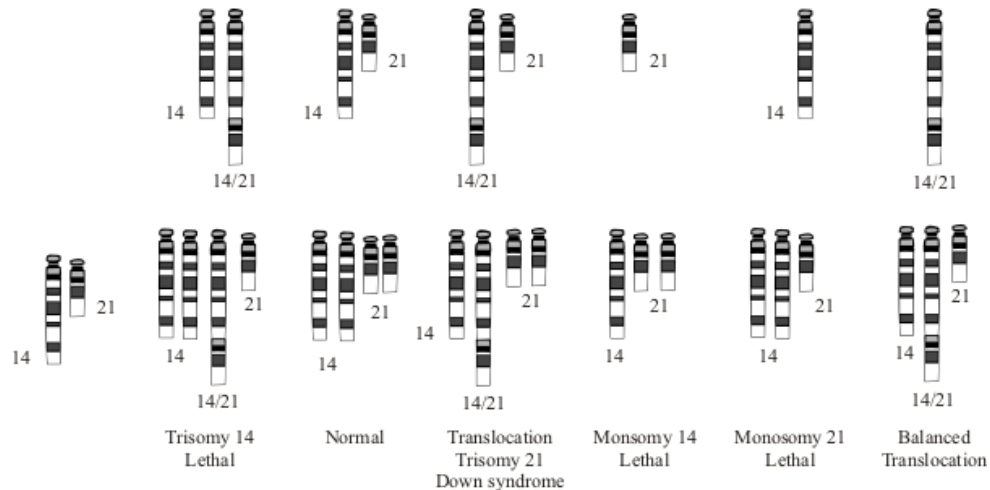
..... [3]

1. The gametes containing chromosomes 14 and 14-21, 14 only, and 21 only will result in embryo mortality / non-viable embryos (OWTTE);
2. Fusion of the gamete containing chromosomes 14-21 and 21 will result in Down's syndrome / trisomy-21;

3. As such, upon fusion with a normal gamete, only the two gametes containing both chromosomes 14 and 21, or 14-21 only will result in a phenotypically normal child;

For reference:

Possible gametes from translocation



Offspring when combined with normal gamete

Upon fertilisation, a zygote contains two copies of chromosome 21 and a fusion chromosome 14-21. All the daughter cells that arise from this zygote have trisomy-21.

- (iii) Explain how events in the cell cycle results in all daughter cells having trisomy-21.

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..... [4]

Points 4+5 compulsory for full marks: 1+2+4+5 OR 1+3+4+5

1. During **S phase of interphase** before mitosis/before prophase, DNA replication occurs to produce genetically-identical sister chromatids via;
2. **Semi-conservative DNA replication** as DNA unwinds and each strand acts as a template for adding nucleotides by complementary base pairing;
3. During **metaphase**, chromosomes are arranged/aligned singly on metaphase plate/at equator;
4. During **anaphase**, genetically identical sister chromatids are separated and move to opposite poles;
5. New daughter cells have same number and kind/type of chromosomes as the zygote, hence all cells will have trisomy-21;

[Total: 10]

Section B

Answer **one** question in this section.

- 4 (a) **Using named examples**, explain the importance of having a three-dimensional structure in different biological polymers. [15]

Carbohydrates – max 3

1. Readily available storage of energy – complex carbohydrates e.g. starch (amylopectin) & glycogen → have many branched ends for faster hydrolysis to release glucose molecules;
2. Compact energy storage – α -glucose chains in amylose, amylopectin, glycogen have helical structures → compact storage as many glucose molecules can be stored per unit volume;
3. Structural fibres – bundling of long, straight cellulose chains to form fibres with high tensile strength → form rigid cell walls and prevents osmotic lysis;

Proteins – max 7

4. Structural proteins – staggered arrangement of tropocollagen molecules minimises points of weakness → form structural proteins with high tensile strength in extracellular matrix;
5. Channel proteins – 3D conformation allows specific molecules, e.g. glucose, ions, to enter/leave cell via facilitated diffusion
6. Cell signalling receptor/relay proteins – extracellular ligand (e.g. insulin, glucagon) must have complementary conformation to bind to specific ligand-binding site of receptors to induce a signal transduction pathway;
7. Cell signalling receptor/relay proteins – binding of relay proteins e.g. G-protein binds to intracellular/cytosolic domain of GPCR to become activated upon change in 3D conformation;
8. DNA-binding proteins – reading of nucleotide sequence of template DNA strand during DNA replication / transcription to produce complementary DNA/RNA strand via complementary base pairing – based on complementary 3D conformation of bases;
9. DNA-binding proteins in eukaryotes – up/down-regulation of expression of specific genes via activators and silencers respectively, binding to specific enhancer and silencer sequences;
10. DNA-binding proteins in prokaryotes – lac/trp repressor protein binding to operator region in *lac/trp* operon;
11. Neutralisation of pathogen/antigen – binding of antibodies to specific epitopes via antigen binding site → prevents binding of pathogen to specific host cell receptors;
12. Opsonisation of pathogen/antigen – binding of Fc portion of antibody to Fc receptors on phagocytes promotes phagocytosis of pathogen/antigen → elimination from body;

Enzymes – max 3

With named example of an enzyme and its substrate / inhibitor:

13. Enzyme-substrate interaction – specific 3D conformation allows for binding of specific substrate molecule (and co-factor/co-enzyme) to form enzyme-substrate complex via lock-and-key mechanism;
14. Enzyme-substrate interaction – flexibility allows for change of 3D conformation in induced-fit mechanism;
15. Enzyme inhibition – specific 3D conformation of competitive/non-competitive/allosteric inhibitor fits active site OR site other than active site OR allosteric site (describe one);
16. After release of products, enzyme can revert to original 3D conformation to be re-used;

Nucleic acids – max 4

17. Nitrogenous bases in polynucleotide strands – 3D conformation of adenine & thymine/uracil, cytosine & guanine → allows for complementary base pairing via formation of hydrogen bonds;

18. Nucleotide sequence in DNA –complementary conformation to DNA-binding proteins → allows for binding of specific proteins (e.g. transcription factors, RNA polymerase) to specific DNA sequences;
19. mRNA – 3D conformation of codons allows for binding of aminoacyl-tRNA complexes via complementary codon-anticodon base pairing;
20. rRNA – specific 3D conformation to fit into active site of aminoacyl-tRNA synthetase for amino acid activation in cytosol (prior to translation at ribosomes);
21. tRNA – has 3D conformation to fit into aminoacyl-tRNA binding sites within ribosome;
22. rRNA – folds into 3D structure to form ribosomal subunits in complex with ribosomal proteins;

QWC – 1

23. ALL three types of biological polymers addressed (carbohydrates, proteins/enzymes, nucleic acids) in a well-organised essay;

- (b) *Shigella* bacteria can cause diarrhoea, fever, abdominal pain. Antibiotics such Ciprofloxacin help treat patients with severe infection or weakened immune system, and reduce the spread of infection by decreasing the number of days the patient has diarrhoea.

However, the number of *Shigella* infections have increased notably since 2013. These strains exhibit resistance to Ciprofloxacin.

Explain how Ciprofloxacin-resistant strains of *Shigella* could have emerged from a single strain of *Shigella*. [10]

Antibiotic Resistance

1. Mutations may have occurred in *Shigella* by random chance, giving rise to Ciprofloxacin resistance allele;

Antibiotic resistance genes can be exchanged between bacteria through mechanisms of horizontal gene transfer (max 6 marks):

Transformation:

2. Fragments of foreign, naked DNA (from lysed antibiotic resistant bacterial cells) in the surrounding medium were taken up by a competent *Shigella* cells via surface proteins;
3. The foreign DNA, containing the antibiotic resistance gene, was incorporated into the *Shigella* chromosome/DNA via crossing over/homologous recombination;
4. If the antibiotic resistance allele were to be expressed in the recipient cell, the bacterial cell had transformed, thus conferring Ciprofloxacin resistance;

Transduction – describe either general transduction or specialised transduction:

5. A phage infected a donor *Shigella* cell, injecting its viral DNA genome into the bacterium;
6. The bacterial DNA was degraded into small fragments, the fragment containing the antibiotic resistance allele was randomly packaged into a capsid head during the spontaneous assembly of new phages;
7. Upon bacterial cell lysis, the defective phage infected another *Shigella* cell and injected bacterial DNA from the previous host cell into the new bacterium;
8. The foreign bacterial DNA replaced the homologous region in the recipient cell's chromosome if crossing over/homologous recombination had taken place, the antibiotic resistant allele from the previous host is now expressed;

Conjugation:

9. Conjugation between a donor F^+ *Shigella* cell and recipient F^- *Shigella* occurred. Sex pilus of F^+ bacterial cell made contact with a F^- cell and retracted to bring the F^- cell closer so a mating bridge is formed between the 2 cells; **R: hollow pilus forms a mating bridge**
10. One of the 2 strands of the plasmid DNA in F^+ cell was nicked and transferred from the F^+ cell to the F^- cell through mating bridge (**R: bridge**) via rolling circle mechanism as the other DNA strand was used as a template for elongation;
11. The single strand F plasmid DNA circularized in F^- cell and was used as a template to synthesise a complementary strand for a double-stranded F plasmid DNA resulting in F^+ cell. The antibiotic resistant allele from the previous host is now expressed in the recipient cell;

Natural selection:

12. In the population of *Shigella*, Ciprofloxacin acts as a selection pressure;
13. Non-resistant bacteria are selected against and bacteria which are resistant to Ciprofloxacin are selected for as they have a selective advantage;
14. The antibiotic resistant *Shigella* cells survive, reproduce and pass down the allele coding for Ciprofloxacin resistance to subsequent generations of bacterial cells during binary fission;
15. Over many generations, frequency of allele coding for Ciprofloxacin resistance increases in the bacterial population resulting in Ciprofloxacin resistant strains of *Shigella*

5 (a) Discuss the causes and significance of variation and diversity in organisms. [15]

Causes of variation:

Mutation (max 3):

1. Gene mutation involves a change in the sequence of nucleotides within a gene and can arise due to substitution, addition, insertion, deletion or inversion (at least 2) of nucleotides.
2. Chromosomal mutations involve a change in chromosomal structure and/or a change in chromosomal number;
3. Changes in chromosome structures could arise due to deletion, inversion, duplication or translocation (at least 2) of chromosome segments;
4. Changes in chromosome numbers arise due to non-disjunction events during meiosis which lead to unequal distribution of chromosomes into daughter cells;
5. A change in the sequence of nucleotides in DNA can result in a change in sequence of amino acids in the resulting polypeptide chain;
6. This could then change the 3D conformation of the protein, affecting its function, and subsequently affecting the characteristics (phenotype) of the organism.
7. When mutations take place during gamete formation, they are inherited/transmitted to offspring and can have an impact on evolution as;
8. these mutations are sources of new alleles in an existing gene pool which contributes to an increase in genetic variation in a population.

Events during meiosis (max 3):

9. Variation also arises due to crossing over between non-sister chromatids of homologous chromosomes at the chiasmata during Prophase I;

10. leads to new allelic combinations in the resulting gametes;
11. Independent assortment of homologous chromosomes during metaphase I;
12. leads to new combinations of maternal and paternal chromosomes in the resulting gametes leading to increased variation.
13. Random fertilization of gametes during sexual reproduction leads to variation amongst the offspring which are produced.

Significance of variation & diversity:

(significance in) Natural selection & speciation

14. individuals within a population show variation for natural selection to act on;
15. those variants with favourable traits have a selective advantage to be selected for and more likely to survive and reproduce;
16. producing viable, fertile offspring, and hence passing on their alleles coding for favorable/advantageous traits to their offspring;
17. proportion of individuals with favorable/advantageous traits increases over time leading to an increase in favourable allele frequencies i.e. microevolution occurs;
18. over hundreds and thousands of generations, evolutionary changes occur and may form new species if reproductive isolation occurs as it is necessary for speciation (i.e. macroevolution occurs);
19. Use of appropriate examples such as Darwin's finches, peppered moths (directional selection by predatory birds after soot kills lichen), long-necked giraffes, resistance in bacteria population (selection in the presence of antibiotics) etc.

(significance in) Heterozygote advantage

20. Variation exists amongst individuals due to the presence of different alleles found in a population. Recessive alleles are preserved in populations due to heterozygote advantage in certain environments;
21. Heterozygote protection maintains a pool of alleles that might not be favoured under the present conditions but some of which could bring new benefits when the environment changes/in a different environment.
22. Ref. Heterozygote individuals with the sickle cell trait are selected for in regions of endemic malaria;

Explanation: HbSHbS homozygotes have sickle cell anaemia, which usually results in early death. Compared to HbAHbS heterozygotes, people with the HbAHbA genotype (normal haemoglobin) have a greater risk of dying from malaria. Thus, sickle cell trait (i.e., the heterozygous condition) is selected for in regions of endemic malaria as the individuals with HbAHbS genotype do not develop sickle cell anaemia and at the same time have less chance of contracting malaria.

(significance in) Frequency dependent selection

23. Variation in populations is also important in the cases where the fitness (hence selective advantage) of the phenotype depends on how common it is;
24. Thus, allele frequencies oscillate/change from time to time depending on environmental conditions. In this case, variation allows for the survival of the species over time;
25. Ref. to example of frequency dependent selection in scale-eating fish;

Diversity of antibodies produced plasma cells

26. Diversity is a key feature of immunological responses by B cells in organisms;
27. (cause) Ref. somatic recombination in naïve B cells. The multiple V, D and J gene segments allow for the generation of many possible distinct antibody molecules, each with unique antibody binding sites.
28. (cause) Ref. to somatic hypermutation in activated B cells - random point mutations in the rearranged VDJ region of heavy chain gene locus and the rearranged VJ region light chain gene locus. By using somatic hypermutation to make changes in the antigen-binding sites of the B Cell Receptors, these receptors and subsequently antibodies produced by differentiated plasma cells, can be “fine-tuned” to have higher affinity for its specific antigen (affinity maturation)
29. (cause) Ref. to class switching in activated B cells - class switching allows for expression of antibodies that have the same antigen specificity (as VH remains unchanged) but have different function (due to different type of CH).
30. (significance): These processes lead to greater diversity of antibodies produced for adaptive immune responses to pathogen infection.

Evolution of viruses – Antigenic shift & Antigenic drift

Qualify that viruses are at ‘the edge of life’ – showing features of both living and non-living characteristics

31. (cause) Variation in viruses is caused by antigenic drift – accumulations of mutations in the viral genome over time leading to changes in 3D conformation of viral proteins (e.g. HA & NA in Influenza viruses) ;
32. These mutations occur due to the lack of proof reading ability of enzymes involved in viral genome replication (e.g. RNA dependent RNA polymerase in Influenza or Reverse Transcriptase in HIV);
33. (cause) Variation in viruses can also result due to antigenic shift – where viruses undergo genetic recombination in a single host to produce new viral subtypes
34. (significance): Viruses are able to overcome host cell immune defence to survive and reproduce better

Variation in bacterial chromosomes – Horizontal Gene Transfer

35. (cause): Ref. to horizontal gene transfer in bacteria.
i.e. Transformation/Transduction/Conjugation
36. (significance): Higher rates of survival and reproduction of bacterial colonies e.g. antibiotic resistance in bacteria.

QWC: ref. to at least 2 different causes for variation and 3 different discussion points on significance of variation

- (b) The human body is capable of responding to changes in the environment. For example, secretion of digestive juices, such as saliva and intestinal juice, occurs in response to ingestion of food. These juices contain enzymes that break down complex carbohydrates into simpler sugars that can be absorbed into the blood.

Describe the cellular changes that occur in response to a meal rich in complex carbohydrates. [10]

Breakdown of complex carbohydrates

1. Exocytosis of hydrolytic enzymes packaged in secretory vesicles to be secreted into mouth/small intestine to break down complex carbs to simpler sugars;

Homeostasis

2. In response to rise in blood glucose concentration above set point of 90 mg/100 mL of blood (A: 90 mg/dL);
3. β cells in islets of Langerhans of pancreas secrete insulin;

Activation of insulin signal transduction pathway

4. Binding of insulin to insulin-binding site of receptor tyrosine kinase, which exists as a linked dimer;
5. Triggers cross-phosphorylation of specific tyrosine residues in the intracellular domain of the other receptor sub-unit;
6. Phosphorylated tyrosine residues in intracellular domain can now be bound by relay proteins, which will then undergo a conformational change to become activated;

Cellular response: uptake of glucose

7. In muscle, liver, and adipose cells, assembly of microtubules for transport of vesicles with glucose transporters / glucose carrier proteins embedded in the vesicle membrane, which then fuses with cell surface membrane;
8. Exocytosis results in an increase in number of glucose transporters / glucose carrier proteins on cell surface membrane;
9. Increase uptake of glucose via facilitated diffusion by muscle/liver cells (AVP: by intestinal epithelial cells);

Cellular response: glycogen synthesis

10. In liver and muscle cells, insulin stimulates the activation of glycogen synthase;
11. To increase the rate of glycogenesis (formation of glycogen) from excess glucose;

Cellular response: triglyceride synthesis and storage

12. In liver and adipose cells, insulin stimulates conversion of excess glucose to acetyl-coA during formation of fatty acids;
13. Fatty acids with a glycerol molecule are then used in the synthesis of triglycerides;
14. Which is then secreted and transported to adipose cells for storage;

Homeostasis: negative feedback loop

15. All these changes act to lower blood glucose concentration back to set point, which when reached, β cells will stop secreting insulin;

QWC – 1

Essay describes chronological description of events following ingestion of carbohydrate-rich meal, with well-organised points into paragraphs.